

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM agents to autonomously decide when to act directly, self-reflect, or consult external knowledge during planning tasks.

arxiv.org/abs/2504.03553

Fast Thinking

Slow Thinking

Know. Thinking

The Problem: Indiscriminate Knowledge Injection in Agent Training

Trajectory Imitation

e.g., ReAct

Blindly imitates gold trajectories without assessing agent need

Trial-and-Error

e.g., Reflexion

Applies external feedback indiscriminately at every decision step

Knowledge-Augmented

e.g., KnowAgent, WKM

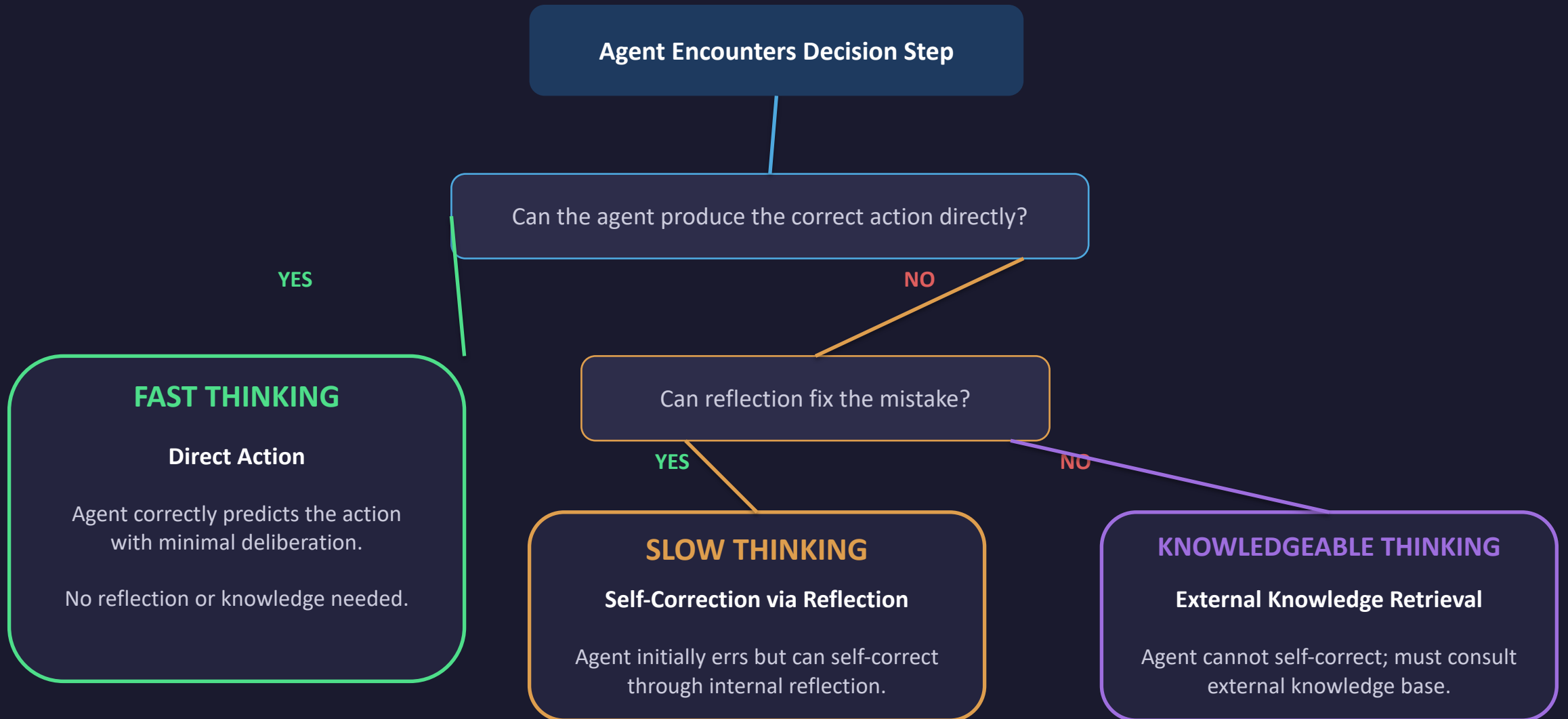
Injects domain knowledge at 100% of steps regardless of necessity

"Flood Irrigation" Consequences: Fragile planning | Pattern collapse on unexpected inputs | Unnecessarily high inference costs

Human Analogy: Situational Self-Awareness

Humans assess whether they can handle a situation directly, need to pause and reflect, or must seek external help. KnowSelf brings this metacognitive capability to LLM agents.

Three Situational Thinking Modes



KnowSelf Framework: Three-Stage Pipeline

STEP 1

Self-Awareness Data Construction

Agent self-explores environments.
At each step, a heuristic criterion
classifies the situation:

Predicted = Gold → Fast
Reflection fixes → Slow
Otherwise → Knowledgeable

Special tokens are inserted to
mark each situation type.

STEP 2

Two-Stage Training

Stage A: Supervised Fine-Tuning
(SFT) teaches the agent initial
self-awareness patterns from
labeled trajectories.

Stage B: Reward-based Policy
Optimization (RPO) further refines
the agent's ability to generate
appropriate special tokens.

STEP 3

Self-Aware Inference

During deployment, the agent
generates special tokens that
trigger one of three actions:

[Fast] → Direct action
[Slow] → Reflection
[Know] → Knowledge retrieval
from external KB

Selective, not indiscriminate.

Qualitative Examples: Where Baselines Fail, KnowSelf Succeeds

Task: "Put a hot mug in cabinet"

OpenAI-O1 (Baseline)

After opening the microwave, O1 incorrectly tries to take an egg from the microwave instead of using it to heat the mug.

Result: FAILS the task

KnowSelf (Llama-8B)

Reflection Token

Uses a [Reflection] token to realize it should heat the mug using the already-open microwave, then proceed with the correct plan.

Result: SUCCEEDS

Task: "Put a saltshaker in drawer"

DeepSeek-R1 (Baseline)

Reflects that the drawer is open and tries to close it first, which fails. Gets stuck in a loop of incorrect actions.

Result: FAILS the task

KnowSelf (Llama-8B)

Knowledge Token

Applies a [Knowledge] token to retrieve the rule: place the object directly without removing unrelated items first.

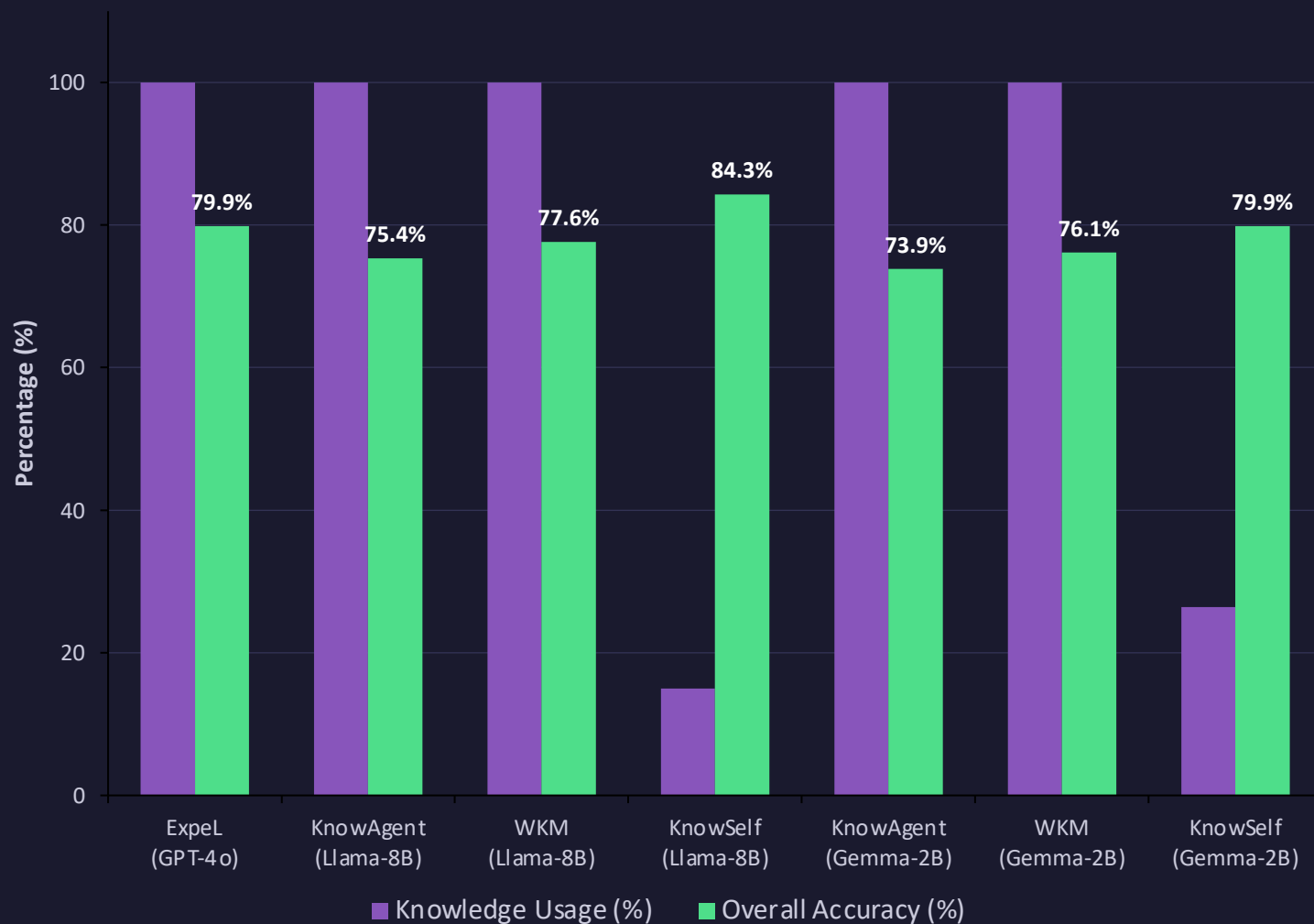
Result: SUCCEEDS

ALFWorld Results: KnowSelf Achieves 84.33% with Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Success rates (%) across six ALFWorld task types. Know% = proportion of steps using external knowledge.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage



Llama-8B + KnowSelf

84.33%

with only 15% knowledge

Gemma-2B + KnowSelf

79.85%

with only 26% knowledge

vs. 100% Knowledge Methods

ExpeL (GPT-4o): 79.85%

KnowAgent (Llama-8B): 75.37%

WKM (Llama-8B): 77.61%

Key Takeaways & Future Directions

1

Indiscriminate injection is suboptimal

100% knowledge methods often underperform selective approaches

2

Agents can learn situational self-awareness

Special tokens + lightweight heuristic labeling, no expensive annotation

3

KnowSelf (Llama-8B) outperforms GPT-4o baselines

84.33% vs. ReAct 76.12% and ExpeL 79.85% on ALFWorld

4

Only 15–26% knowledge usage needed

Dramatically lower inference cost vs. 100% for KnowAgent/WKM/ExpeL

5

Scales across model sizes

Even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL

Future Directions



Extend the three-situation framework to multi-agent collaborative settings



Investigate learned self-awareness on more complex benchmarks beyond ALFWorld



Explore dynamic knowledge base construction and curation strategies



Study transfer of self-awareness capabilities across different domains and tasks



Combine with retrieval-augmented generation for richer knowledge integration